

# Distance between a line and point

## Key Points

The (shortest) distance between a line and a point is the **perpendicular distance** between the line and the point

Find  $\lambda$  such that  $(\mathbf{r} - \mathbf{p}) \cdot \mathbf{d} = 0$  to find the point  $\mathbf{r}$  on the line perpendicular. Then the distance is  $|\mathbf{r} - \mathbf{p}|$

## Basic Skills

**Q1.** Find the distance between the following coordinates

a)  $A=(0,6)$ ,  $B=(-3,10)$       b)  $A=(5,6,2)$ ,  $B=(-1,-2,3)$

**Q2.** Given  $\mathbf{a} = \begin{pmatrix} -2 \\ 5 \\ 0 \end{pmatrix}$  and  $\mathbf{b} = \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$

- a) Find  $|\mathbf{b} - \mathbf{a}|$
- b) By explaining what  $|\mathbf{b} - \mathbf{a}|$  represents geometrically, explain why  $|\mathbf{b} - \mathbf{a}| = |\mathbf{a} - \mathbf{b}|$
- c) Compute  $\mathbf{a} \cdot \mathbf{b}$  and hence find the angle between  $\mathbf{a}$  and  $\mathbf{b}$
- d) Find a unit vector for the direction of A to B
- e) Write the equation of the line through AB

## Competency Skills

**Q3.** A line  $l$  has the equation  $\mathbf{r} = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$  and P is the point  $(3,4,5)$

- a) Confirm if point P is on the line or not
- b) By writing  $\mathbf{r}$  as a single vector compute  $(\mathbf{r} - \mathbf{p}) \cdot \mathbf{d}$
- c) Hence find the point on the line perpendicular to point P
- d) Hence find the distance between the line  $l$  and the point P

**Q4.** Find the distance between the line  $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$  and the coordinate  $(2,5,1)$

**Q5.** Find the point on the line  $\frac{x-2}{3} = \frac{y}{4} = \frac{z+3}{-1}$  and the point with position vector  $3\mathbf{i} - 10\mathbf{j} + \mathbf{k}$

## Advanced Skills

**Q6.** Find the distance between the line  $x = y = z$  and any coordinate

**Q7.** Find the constant  $\alpha$  such that point  $A = \begin{pmatrix} \alpha \\ -1 \\ 1 \end{pmatrix}$  and the line  $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$  are the minimal distance away

**Q8.** Find the distance between the parallel lines  $\mathbf{r}_1 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}$  and  $\mathbf{r}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -9 \\ -3 \\ 6 \end{pmatrix}$  by fixing a point on one of the lines

## Extension

**Q9.** Can we fix a point to work out the distance between two non-parallel lines

Hint: Try it where you move the point and see if the answer changes